

Low-Frequency Electronic Muscle Stimulator

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Here is a simple low-frequency electronic muscle stimulator used in electrotherapy (use of electrical energy for medical treatment).

Electronic muscle stimulation or electronic stimulation makes use of a small voltage that aims at the motor nerves in a human body to excite these. It basically causes contraction of muscles. The muscles rest between shocks and contract again when the shocks occur. This regular contracting and relaxing has been used to cure various vascular and muscu-

loskeletal conditions. This method makes the movement of oxygen and nutrients to the muscles much easier. General applications of electronic muscle stimulation are to stimulate sensory and motor nerves, facilitate muscle contraction, re-educate muscles, improve circulation and prevent/stretch adhesions.

Circuit and working

Fig. 1 shows block diagram of the muscle stimulator. The power-supply block has a transformer, 230V AC mains as input, bridge rectifier and filter.

The variable-voltage block provides variable voltage to the converter block. It is built around IC LM317 and produces voltages of 1.25V to 25V with a current of 1A. This block produces the required power for upping the voltage to the

required level.

The fixed-voltage block provides a voltage of 5V to the astable multivibrator block.

The astable-multivibrator block is built around IC LM555 and produces low-frequency variable square wave. It provides frequency to the converter block, which produces the signal for the output block. Output from the converter is given to various points on the human body for stimulating the muscles.

Fig. 2 shows circuit diagram of the stimulator circuit. It is built around transformer X1 and a bridge rectifier comprising diodes D1 through D4, filter capacitors C1 and C2, 5V voltage regulator 7805 (IC1), adjustable regulator LM317 (IC2), timer LM555 (IC3), transformer X2 and a few discrete components.

IC1 produces 5V fixed voltage to enable IC3, and IC2 produces adjustable voltage which is given to X2. LED1 indicates the presence of 5V DC.

The 555 timer is configured in astable-multivibrator mode. For stimulating muscles, low frequencies are used, ranging from 0.7Hz to 31Hz, which can be varied using potmeter VR1.

Output frequency at pin 3 of the 555 timer is used to drive transistor

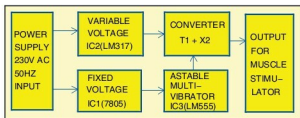


Fig. 1: Block diagram of the muscle stimulator

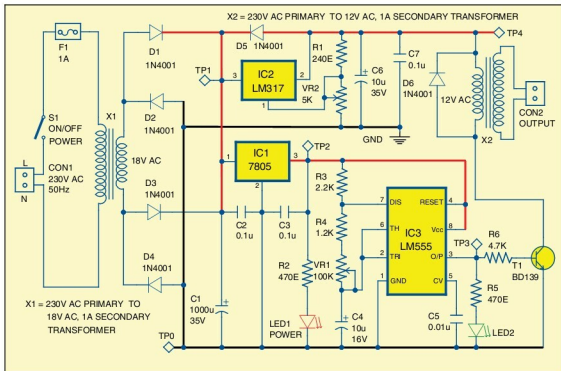


Fig. 2: Circuit diagram of the muscle stimulator

